

Digitizing Loan Approvals for Financial Inclusion

Using Machine Learning to Power Smarter Credit for Kenyan MFIs & Digital Lenders

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Kenya's Financial Landscape Challenge

MFIs: Traditional Lifeline

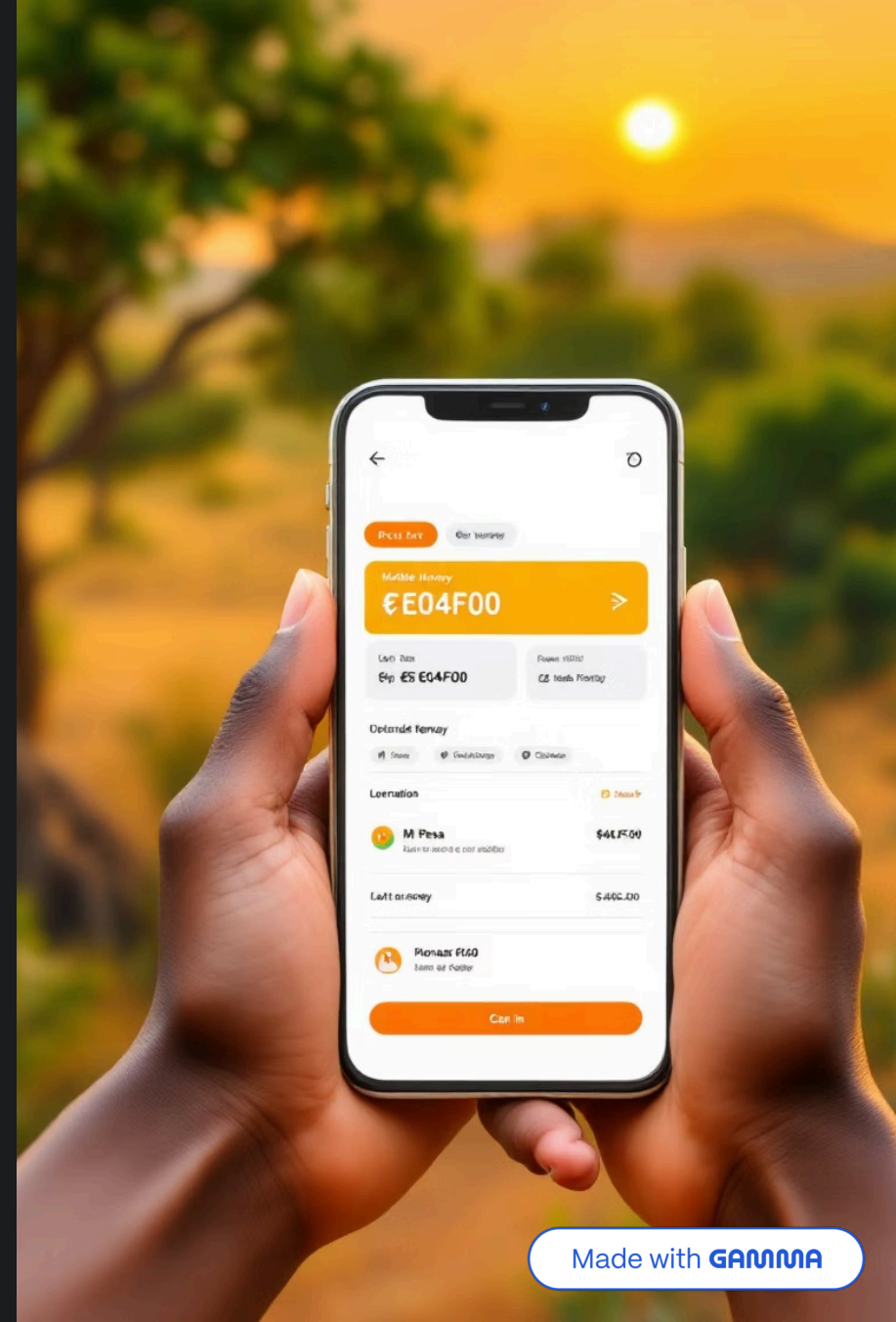
Serve underserved communities, women, and small businesses but face slow manual approvals and limited digitization

Digital Lenders: Tech-Powered

Instant AI-driven approvals via M-Pesa integration, but struggle with data privacy and rural access

Our Solution

Bridge the gap with machine learning models for faster, fairer, and more efficient credit decisions



Stakeholders & Value Proposition



MFIs

Must digitize to cut costs, reduce delays, and stay competitive in evolving market



Digital Lenders

Provide instant credit but face privacy, security, and inclusion gaps



Applicants

Women, rural entrepreneurs, and small & medium enterprises (SME) need faster and fairer credit access



Regulators

Demand transparency, responsible lending, and sustainable growth

Value Proposition

- Automated approvals that reduce bureaucracy
- Fairer, data-driven credit scoring for underserved borrowers
- Stronger compliance and consumer protection for regulators

Objective

Deliver a supervised ML model for automated, fair, and efficient loan approvals, supporting Kenya's transition to a more inclusive and resilient financial sector

Data Overview

Dataset Foundation

Loan applications with financial, demographic, and outcome details

Target variable: loan_status (Approved vs. Not Approved)

Key Features

- **Loan Features:** loan_amnt (amount), term (duration), int_rate (interest rate)

Captures the core structure of a loan; critical for assessing affordability, risk exposure, and repayment likelihood

- **Credit Indicators:** grade / sub_grade

Provides proxy measures of borrower creditworthiness, directly influencing default probability

- **Applicant Information:** emp_length, home_ownership, annual_inc (income)

Reflects borrower stability and capacity to repay, key for fair and data-driven approvals

- **Purpose of Loan:** purpose (e.g., debt consolidation, credit card, home improvement)

Certain purposes correlate strongly with risk patterns, helping lenders adjust approval and pricing strategies

- **Outcome:** loan_status (target)

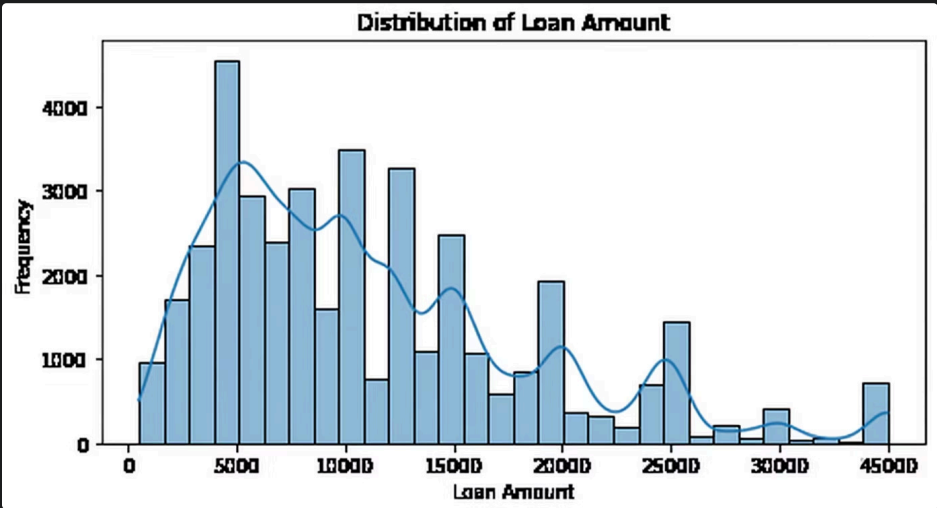
Defines the success of the prediction task; the benchmark against which model performance is evaluated



Data Insights from EDA

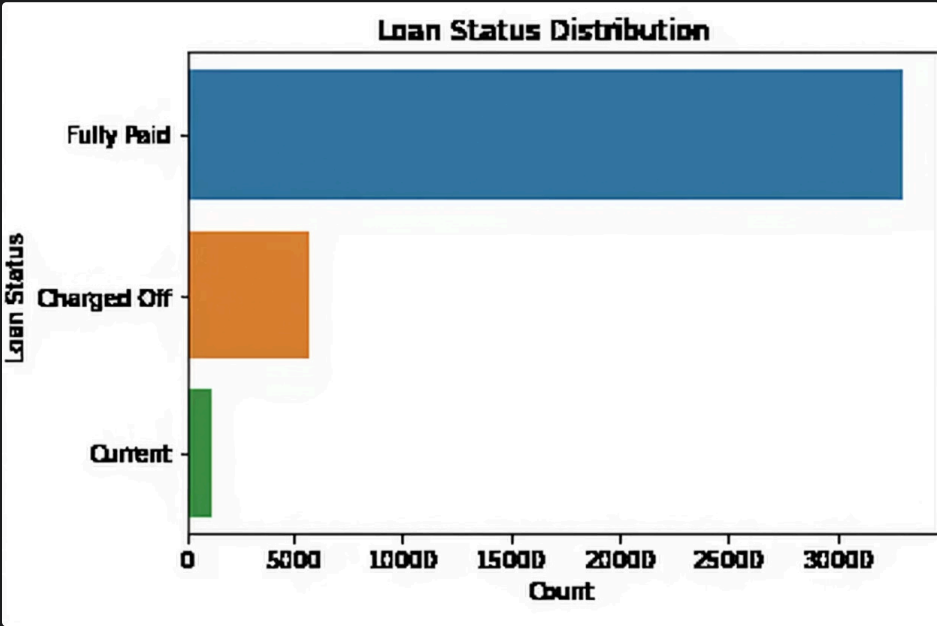
Loan Amounts

Most borrowers request modest loans with few large outliers. Guides lenders in setting loan limits



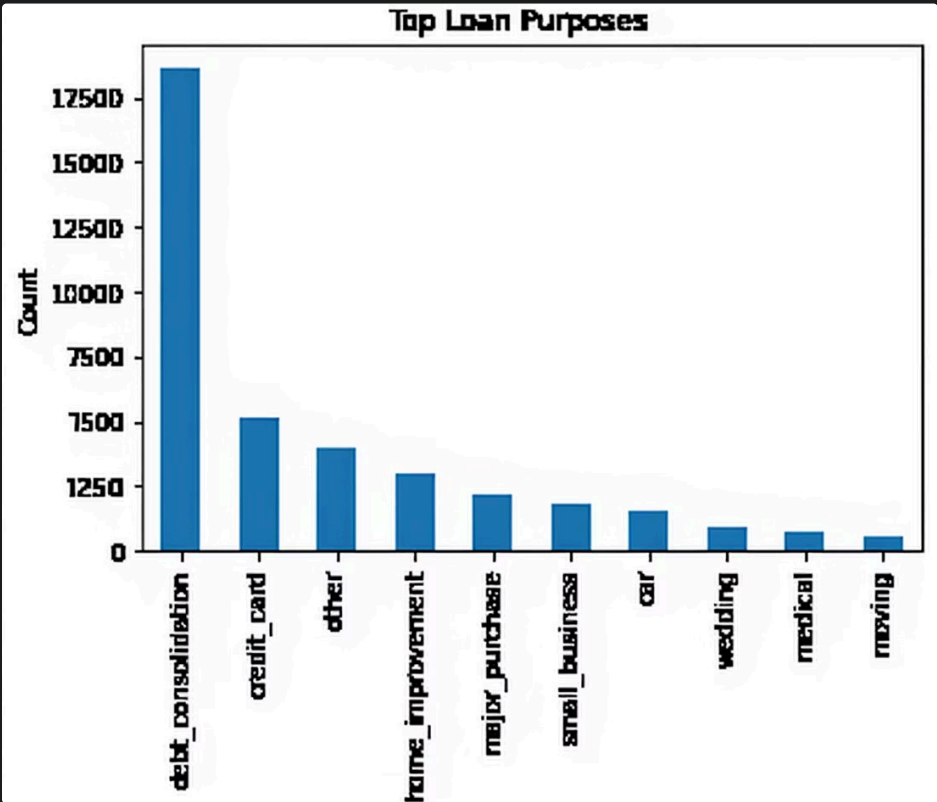
Loan Outcomes

Majority fully paid; defaults rare but high impact
Confirms focus on PR-AUC & recall metrics



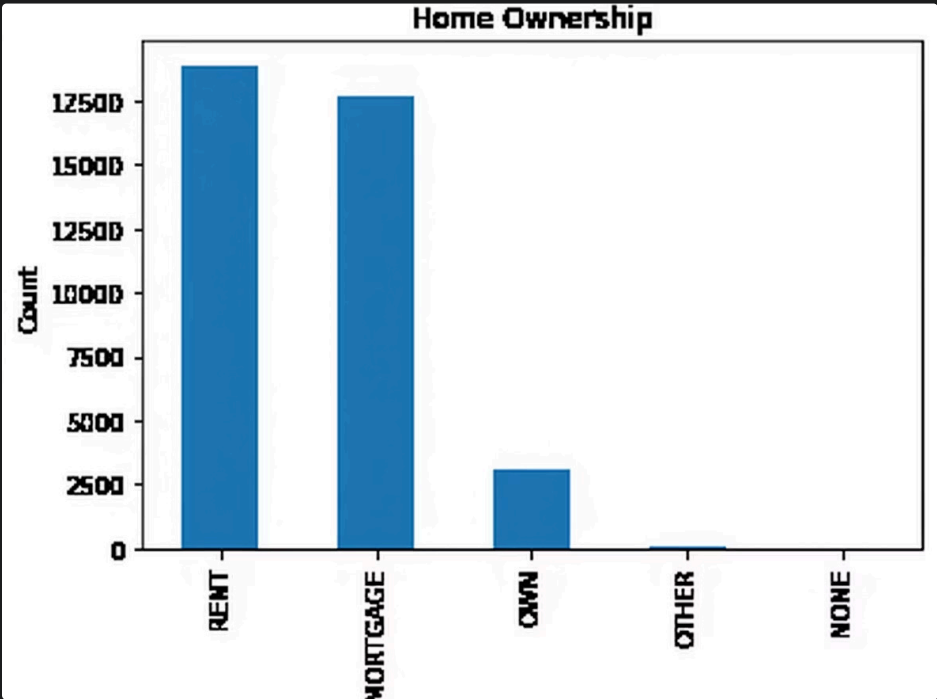
Loan Purposes

Debt consolidation dominates, followed by credit cards
Certain purposes signal higher risk



Home Ownership

Borrowers mostly rent or have mortgages
Ownership type predicts repayment ability



Modeling Approach

We tested four machine learning models to automate loan approvals, each bringing unique strengths:

01

Logistic Regression

Statistical baseline predicting probabilities. Easy to interpret and useful benchmark for complex models

02

Random Forest

Builds hundreds of decision trees and averages their results to reduce bias and variance, capturing non-linear patterns. More accurate than Logistic Regression, but less transparent and sometimes unstable

03

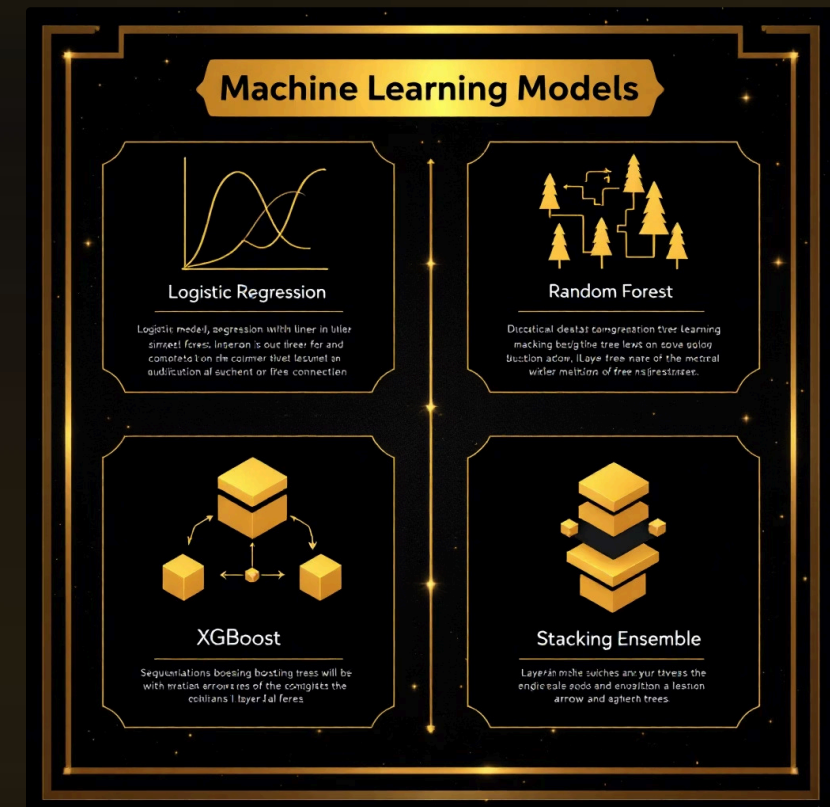
XGBoost

A gradient boosting method that builds trees sequentially, each correcting the errors of the previous ones. Handles missing values and outliers well. Delivers strongest precision-recall balance

04

Stacking Ensemble

Combines Random Forest + XGBoost models. Slight improvement but added complexity without major business benefit



Key Performance Metrics



PR-AUC

Precision-Recall Area Under Curve focuses on correctly identifying approved loans. Crucial for imbalanced data and aligns with the business goal of maximizing fair approvals while minimizing risk



Recall

Measures actual approved loans correctly identified. High recall means fewer missed opportunities for eligible applicants



Precision

Measures how many predicted approvals are actually correct. High precision reduces the risk of lending to risky applicants



F1 Score

Balances precision and recall, providing single measure of overall model quality

Model Performance Results

We compared four models on the holdout test set using key metrics:

Model	PR-AUC	Precision	Recall	F1 Score	Interpretation
Logistic Regression	0.897	0.889	0.573	0.697	Simple but missed defaults thus higher risk of losses
Random Forest	0.913	0.864	0.832	0.847	Better balance than baseline but less stable
XGBoost	0.915	0.863	0.871	0.867	Strongest balance: approves more good borrowers while controlling risk
Stacking Ensemble	0.911	0.848	0.917	0.881	Marginal gains but more complex to deploy & maintain

Why XGBoost Wins



Best Performance

Highest PR-AUC (0.915) → most suitable metric for rare defaults in loan data



Balanced Risk

High recall (0.871) → fewer deserving applicants wrongly rejected, allowing more eligible borrowers to access credit responsibly

High precision (0.863) → fewer risky loans approved, reducing default risk and improving overall portfolio quality



Imbalanced Data Handling

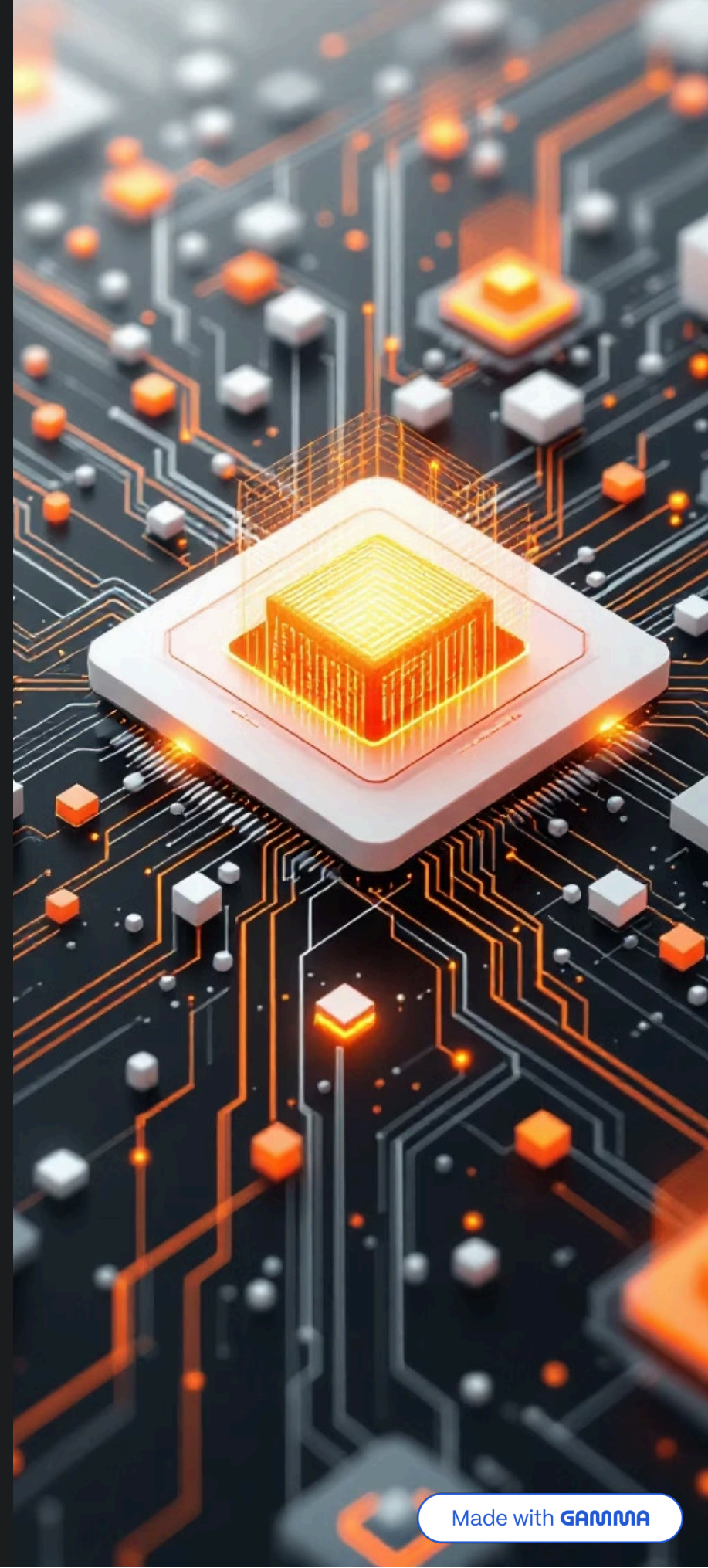
Focuses on learning from rare default cases, maintaining high precision and recall even when approved loans greatly outnumber defaults—making it more dependable than models that rely on accuracy alone



Business Ready

Robust to outliers, handles missing data, fast processing, & provides clear feature importances

XGBoost outperformed all models making it the most effective and practical model for fair, efficient, and scalable loan approvals delivering real financial inclusion



Integrating Sentiment Analysis with XGBoost

To further enhance our loan approval process, we are integrating sentiment analysis, providing a deeper understanding of applicant intent and tone

Overview

Borrower descriptions are analyzed using VADER sentiment analysis to capture tone and intent, adding a sentiment score to the XGBoost model for more informed decisions

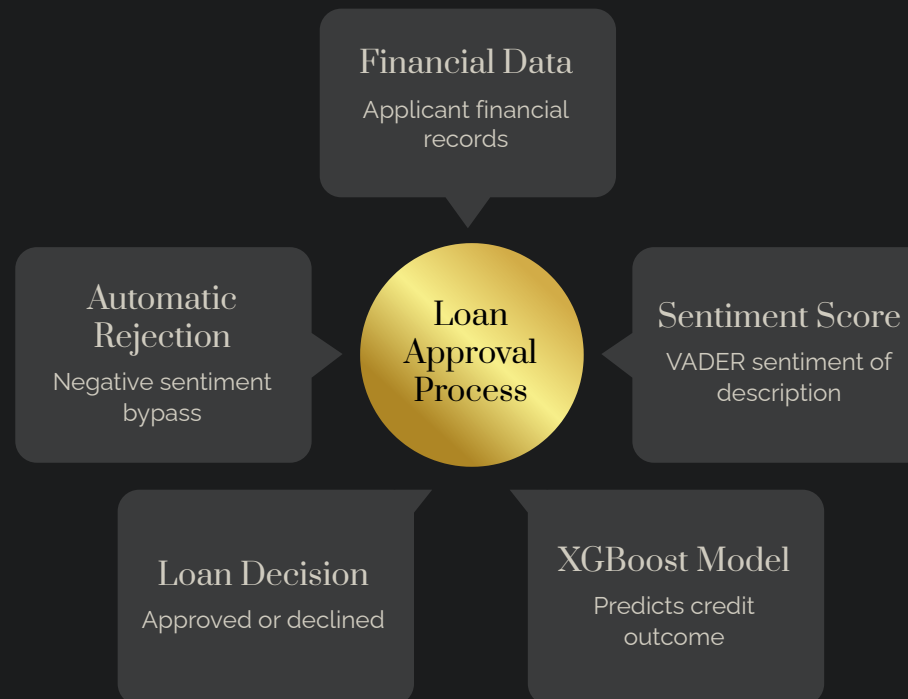
Sentiment-Based Loan Rule

If a borrower's sentiment score is too negative, the application is automatically flagged or rejected. Otherwise, the model evaluates the loan as usual

Business Impact

Adding sentiment improves risk detection and ensures fair, responsible lending decisions by considering both quantitative data and qualitative borrower behavior

This flow diagram illustrates how sentiment analysis is integrated into our enhanced loan approval process:



Recommendations

Business Value

- Identifies ~87% of eligible applicants → fewer deserving rejections
- 86% precision → lowers lender's exposure to bad loans
- Strong PR-AUC → consistent performance across thresholds

Operational Recommendations

- Deploy in decision pipeline with feature transformations
- Start conservative, A/B test for trade-offs
- Monitor drift & fairness monthly
- Retrain regularly with governance checklist

Next Steps

- Retrain regularly & adopt a model governance checklist
- Integrate additional data sources (e.g., mobile transactions or credit bureau data) to enhance prediction accuracy and inclusiveness



Empowering financial inclusion through faster, fairer, and smarter lending 🙏

Thank you

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💡 *"Together, we're shaping the future of inclusive finance in Kenya — one intelligent, data-driven loan decision at a time."*

